Student ability estimation based on IRT

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***Abstract*—Most of the assessment systems are now using the Classical Test Theory (CTT), the real ability of students is not exactly revealed because they rely only on counting the number of true responses without awaring other characteristics like the difficulty of each item. Several testing software are applied weighted questions but they depend on the sentiment of teachers.**
The modern testing theories nowadays are built on a mathematical model which can calculate the latent trait of students. The Rasch model is the probability model which promotes interaction between an item and a student.
We have constructed a system to estimate students' ability basing on Item Response Theory (IRT) and applying K-Means to classify student ranking. In this paper we present a model to categorize students' levels and compare them to traditional assessment methods. The results indicate that the methods we proposed have shown some significant improvement and they could be effectively applied for other tutoring systems. The result is also meaningful in customizing content and testing ways. Beside, the research is a guide line for teachers or test makers to give other testing approaches for various examinations.

***Index Terms*—Item Response Theory, Students' Ability Estimation, Maximum Likelihood Estimation, Item parameters estimation, k-Means.**

I. INTRODUCTION

Current classical assessing tools for multiple-choice test are designing to count the number of correct answers on the total number of questions that students must answer. Other systems have been built with weighted questions. The construction of the current weighted question depends solely on the teacher's subjective that reduces objectivity in evaluating the ability of candidates.

Although the modern testing theory was founded from the 50's of the last century but until at the end of 70's and 80's it was widely used [3]. The reason was that it received strong support from personal computers and number of psychological scientists, especially psychometricians.

Applying IRT for assessment based on Computerized Adaptive Testing (CAT) is widely developed. In fact, there are not many completed testing systems using IRT on the background. Most of these systems are using CTT which is easy to meet test data [8].

IRT model provides a framework for assessing the ability of each student and the difficulty of test item instead of the test like CTT [8]. It can be named *hidden latent model* where the latent is discovered by observing variables and properties or parameters of the models.

In recent years, many applications using IRT have concentrated on the estimation of students' ability to give suggestions to adapt learner's tutorial or documentation. Many e-learning and assessment systems based on IRT are mainly oriented to ability estimation in order to suggest adjusting learning content or change test difficulty level in the personalized learning systems [15]. Others applications firstly applied IRT for capability estimation and then used classification methods for student ranking [2] [20].

In the USA, some of the well-known testing systems are GRE (Graduate Record Examination)(http://www.gre.org) and TOEFL (Test Of English As a Foreign Language) (http://www.toefl.org) which are using IRT in student assessment.

This study focuses on applying IRT to replace traditional testing method to improve accuracy in comparing student's ability. After that, k-Means is conducted to cluster student levels into 10 groups. At this stage, we suppose to use median method instead of mean in k-Means algorithm for better distribution curve which looks like normal distribution one. Following these classifications, we can rank exactly the real performance of each student groups and identify the ability of all students in the test.

II. THEORY BACKGROUND

IRT model is based on three assumptions:

1) Ability is a scalar value θ;
2) All the items are independent;
3) The answer of a student to a question can be described by a mathematical function.

According to the theory of IRT, there are two concepts:

1) The possibility of one student who answers correctly a question can be calculated by a set of factors and parameters;
2) The relation between the possibility of a correct answer and the set of factors can be described by a continuous increasing function called item characteristic curve.

A. Ability estimation theory

We assume that each person has an ability scale called θ. For each ability level, there is a possibility of a correct answer called $P(\theta)$. This possibility keeps small value if student's ability is low and vice-versa. The graph representing for θ and $P(\theta)$ value has S shape. There are three specific IRT models which are classified and based on the number of parameters. They are one-parameter model, two-parameters model and three-parameters model.

- One-parameter model: This model is represented in a mathematical equation:

$$P(\theta) = \frac{1}{1 + e^{-(\theta - b)}} \tag{1}$$

  Where: θ is ability level, b is item difficulty parameter. Difficulty parameter is ability mark to get correct answer which possibility is 0.5, the popular level of this parameter is between $-3 \leq b \leq +3$ $P(\theta)$ is the possibility of getting correct answer with θ ability.
- Two-parameter model: This model add one more parameter: discrimination. The discrimination can be called the slope of item specification curve, other systems consider this parameter as a constant $a = 1.702$. In our system, $a = 1$ parameter is calculated by division between 27% upper students and 27% lower students who answer items correctly [12].

$$P(\theta) = \frac{1}{1 + e^{-a(\theta - b)}} \tag{2}$$

- Three-parameter model: This model includes guessing parameter based on property: In order to test, student can answer correctly by guessing without understanding the item.

$$P(\theta) = c + (1 - c)\frac{1}{1 + e^{-a(\theta - b)}} \tag{3}$$

  Generally, c parameter received value between $0 \leq c \leq 0.35$ [3]

Almost applications estimate students' ability basing on two-parameter model, the other apply one-parameter - Rasch model or three-parameter model (including guessing parameter) [6].

When estimating students's ability using IRT method, depends not only the number of correct answers but also each item characteristics. It is the fact that if two students correctly answer the same items they must receive the same result. However, if two students answer correctly the same number of question but not the same items, the results can be different. This is the different point between two estimation model CTT and IRT (they can be called linear and nonlinear model, respectively). These also are reasons for us to choose two-parameters instead of one-parameter because one-parameter model sets default for all items with the same difficulty = 1. According to Baker [3], there are three methods to estimate ability:

- Ability estimation with clear question parameters: This is the easiest way, ability estimation process can begin with an ability initialized values. This value is used to calculate the probability of correctly answer questions. After that it can be modified to improve the calculated probability value to fit to the results of answered questions. The process continues until the value of adjustment is smaller than a threshold value and estimated ability is not significantly changed. This process is repeated for all of the students participating in the test.
- Question parameters estimation with clear student's ability: It is similar to the ability estimation process above. First, we initialize the default value for each item parameter, then we calculate the $P(\theta)$ value. The blend of calculated probability values and observed probability values may be determined by calculating ability scales. These processes can be continued until the changed values are acceptable, in experiments, these values are smaller than 0.0001.
- Ability and question parameters estimation: estimating both two values in the same period of time may be more sophisticated. It is the combination of two contemporaneous processes: Generally, the first step is started by default pre-values to estimate ability value. Then, the collecting value is used to estimate question parameters. This process is repeated until the adjusted parameters change insignificantly (compare to item characteristic curve model (ICC) and calculated probability).

B. Estimation methods

To obtain more precise estimation of learners' ability, many procedures are implemented. These alternative procedures for estimating IRT parameters and to be employed to determine the ability of examinees as they are submitting their answers to each test item. There are three basic estimation methods that are used in IRT for parameters estimation:

- Student's ability estimation using maximum likelihood estimation method - MLE. Ronald A. Fisher introduced firstly in 1912 [1] [11].
- Student's ability estimation using Bayes Modal Estimation method - BME [18].
- Student's ability estimation using Weighted Likelihood Estimate - WLE, proposed in 1989 by

Thomas A.Warm for reducing bias in estimation [19].

Current widely use the maximum likelihood estimation (MLE) is applied to estimate the learner ability. In our experiment, MLE method was chosen because of test results are good enough for small test (not too many questions in the test). Bayesian methods might issue higher estimated error rate [11, page 148].

C. MLE Method

Students' ability estimation with Maximum likelihood estimation algorithm uses likelihood estimation function [11, page 33-40]

$$L(u_1, u_2, ..., u_n|\theta) = \prod_{j=1}^{n} P_j(\theta)^{u_j} Q_j(\theta)^{1-u_j} \quad (4)$$

where:

$$P_j(\theta) = \frac{e^{(\theta - b_j)}}{1 + e^{(\theta - b_j)}}$$

and $Q_j(\theta) = 1 - P_j(\theta), P_j(\theta)$ is probability of the tester who answers the question j correctly with $\theta, Q_j(\theta)$ is probability of incorrect response. $u_1, u_2, ..., u_n$ are corresponding values to true/false response. $u_j = 1$ if it's true and $u_1 = 0$ if vice-versa.

Student's ability is identified by the maximum value of estimation function [11].

Information function conducts the comparison between testers and item difficulty (this function checks the accuracy of estimation process, in another way, it gives us the precision of estimating learner's ability). The information function does not depend upon the distribution of examinees over the ability scale [3].

$$I_j(\theta) = a_j^2 P_j(\theta) Q_j(\theta) \quad (5)$$

There are two special cases which MLE function cannot identify ability values, these happen when students answer all items correctly or wrongly. The ability estimation is positive $(+\infty)$ and negative infinity $(-\infty)$ respectively [3, page 89]. In both of these cases, when the value $\Delta\theta$ does not converge smaller than 0.001, the algorithm will be stopped [20].

D. K-Means algorithm

According to estimated result, $k-$Means method was used to cluster students into current popular accreditation levels. Many grading systems all over the world choose 10 levels corresponding to 10-point scale or 100-points scale systems.

$K-$Means was firstly introduced by Jame Macqueen in 1967 [16] through the idea of Hugo Steinhaus [9] created 10 years before. $K-$Means is an important algorithm and is widely used in clustering techniques. The main idea of $k-$Means is finding a method to partition all objects into k clusters ($k-$ is the number of clusters, k is a positive integer number) in which each observation belongs to cluster with the minimum of sum of distance functions of each point in the cluster to the centroid.

The $k-$Means algorithm is conducted following these steps:

1) Select randomly k centroids for k clusters.
2) Calculate the distance between all objects to k controids (Euclidean distance).
3) Group objects together with minimize distance.
4) Reallocate new centroids by summarizing all distances to each centroid.
5) Repeat step 2 until nothing change between clusters.

III. Related work

With the rapidly developing of computers and educational technology, e-learning and tutoring systems have become more and more important. Their appearance gradually replaces other traditional learning. Under the supporting of computers, many intelligent systems takes part in training and assessing process. The combination between computer science and psychology is also noticeable. Many researchers concentrate on appling psychology into intelligent tutoring systems for efficiently decision systems. Young-Jin Lee developed useful computational method to estimate the ability of students in a Web-based learning environment [14]. In 2009, Wen-Chih Chang proposed the combination between IRT and K-means to estimate and diagnose issue in e-learning environment [20]. They did experiment with two groups, the first one did the test without feedback and comparing with the second one with full supportive feedback. Comparing the test result showed the better performance for the second group. Chen et al at Graduate Institute of Learning Technology, Taiwan has introduced an e-learning system using IRT for personalizing course material [4] [5]. Other authors deploy a neutral network for personalization, predictation in chosing appropriately target courses [7].

It is the fact that, there are many studies about IRT and clustering algorithms, but they are conducted separately. To join these ideas and make it useful in a lifestyle application which supports makers' decision to know the exact performance of students. This is the reason why our research was focused on.

IV. Experiment

The experiment was conducted on the testing data of the entrance examination to Gifted school - Hanoi National University of Education. It included English testing responses of 1111 examinees taken into grade 10 in 2014. The output could be compared to true score to realize the differences between the ways of performance sort. Through this experiment we proposed a new ranking and accreditation method to help the administrators to give accurate decisions for this test. This modul was implemented on Python version 3.4 (http://www.python.org).

A. Implementation

- Index test result in ascending order of raw scores.
- P_H denoted 25 percent of students getting the highest score. P_L is 25 percent of students getting the lowest scores. Counting number of correct responses and how many percents in this group of each question (According Kelly [12]).
- Calculate the initial difficulty of each question $b = rawscore =$ *correct response/all question.*
- Calculate discrimination for each question: $D = P_H - P_L/$*(total samples).*
- Initialize the first ability value θ for each student, in our case it is 1.
- Calculate probabilities $P(\theta)$ and $Q(\theta)$
- Repeat to calculate above step until $\Delta\theta$ value changed insignificantly. Popularly, in many experiments it is $\Delta\theta \leq 0.0001$ [3]

$$\theta_{s+1} = \theta_s + \frac{\sum_{i=1}^{N} -a_i[u_i - P(\theta_s)]}{\sum_{i=1}^{N} a_i^2 P_i(\theta_s) Q_i(\theta_s)} \quad (6)$$

- Applying $k-$Means to cluster in which k is standard scale.

B. Experiment result

Input data is included in 2 tables "Response" and "Key". The first one is a set of 1111 records of examinees in which are standardized to ABCD for 50 questions. The last one showed the key for each question.

StudentID	Response
1	CACBDACDCDBAAD...CDBBAABCBCCABD
2	CAABDACACDABBC...CADCABBDDCCADA
3	CACBDACACDABBD...CDAABCDDDCCADD
4	CACBDACACDACAD...CDBCDCBCCCCADD
5	CACBDACACDABCD...CDBAABDCDCCADD
6	CACBDABACDABBD...DDDADCDDDCCADD
7	CACBDABACDDCAD...CDBADCBDDCCAAA
8	CACBDACACDABBD...CDAADCDDDCCADD
9	CACBDACACDABBD...CDDADCBDCCCADD
10	CACBDACACDABBD...CDADACDCDCCADD
11	CACBDACACDABBD...CDDAACDDDCCADD
12	CACBDACDCDABBD...CDDADCDCDCCADD
13	CACBDACDCDACBD...CDDADCDDDCCADA
14	CACBDACCCDABBD...CDAADDDDDCCADD
15	CACBDACACDACBD...CDDBDCBDDCCADD
16	CACBDACACDADAD...CDBADCBDACCADA
17	CACBDACACDABBD...CDAAACDDDCCADD
18	CACBDACACDABBA...DDDADCDDDCCADA
19	CACBDACACDABBD...CDDAACDDDCCADA
20	CABBDACACDABBA...CDABCCDDCCCAAD

TABLE I: The answers of students

The student's answers will be compared to the keys table to get u_i value, it is 1 if it matches and is 0 if vice-versa.

item	Key
1	C
2	A
3	C
4	B
5	D
6	A
7	C
8	A
9	C
10	D
11	A
12	B
13	B
14	D
15	B
16	B
17	C
18	A
19	B
20	B
21	...

TABLE II: Keys for question

The next table shows the result of estimating process, these intermediate values are collected after 28 iterations (number of iterations are depended on the convergent of θ):

item	u	P	Q	a(u-P)	a^2PQ
1	0	0.3857	0.6143	-0.2335	0.0868
2	0	0.401	0.599	-0.1904	0.0542
3	0	0.4177	0.5823	-0.169	0.0398
4	0	0.4411	0.5589	-0.1165	0.0172
5	1	0.4181	0.5819	0.218	0.0341
6	1	0.4312	0.5688	0.175	0.0232
7	0	0.4517	0.5483	-0.0967	0.0113
8	1	0.4203	0.5797	0.3199	0.0742
9	1	0.413	0.587	0.2454	0.0424
10	1	0.4417	0.5583	0.1456	0.0168
...	...	...	...	...	...
40	0	0.4109	0.5891	-0.2268	0.0737
41	1	0.4976	0.5024	0.0185	0.0003
42	0	0.3895	0.6105	-0.2462	0.095
43	0	0.4368	0.5632	-0.2688	0.0932
44	0	0.3957	0.6043	-0.2515	0.0966
45	1	0.4033	0.5967	0.3732	0.0941
46	1	0.4141	0.5859	0.2352	0.0391
47	0	0.4444	0.5556	-0.107	0.0143
48	1	0.4093	0.5907	0.2489	0.0429
49	1	0.4066	0.5934	0.3116	0.0665
50	1	0.4204	0.5796	0.3315	0.0797

TABLE III: Estimate output of maximum likelihood function for one student

The following figure shows the distribution of θ:

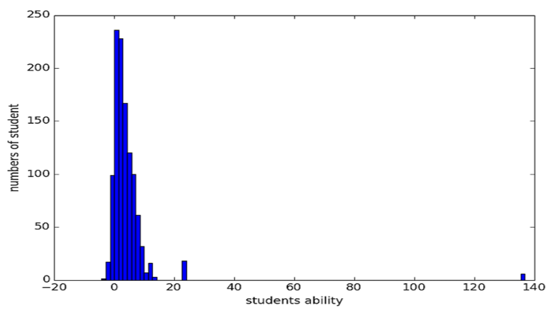

Fig. 1: Histogram of student abilities

Comparing to raw score of CTT method - the observed scores:

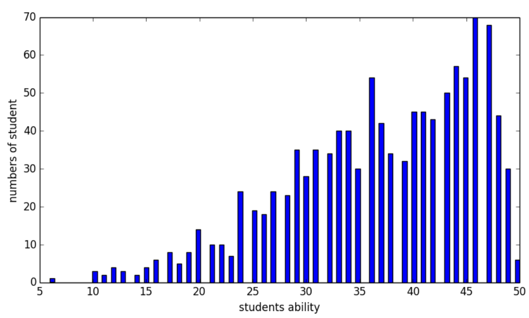

Fig. 2: Student's ability distribution

V. Experiment discussion

After applying IRT, the range of θ is more detailed than raw score (θ are smoother excepted for outliers. Referring to the table III in page 4 for further). It helps us more easily classify students in the sense of higher differentiation. At the next step, we use k-Means to cluster these θ values.

In the k-Means method, if we calculate mean value, we will receive lower accuracy because of the outlier values. To get more accuracy, we propose to use median alternatively to eliminate noise problem [17, page 200]. The comparison between two methods is shown below:

Comparing mean and median						
k	Median			Means		
	Centroid	Count	Rate%	Centroid	Count	Rate%
1	-1.989	18	1.62	-1.163	66	5.94
2	-0.574	78	7.02	0.263	153	13.77
3	0.450	138	12.42	1.367	209	18.81
4	1.324	165	14.85	2.508	184	16.56
5	2.240	164	14.76	3.889	170	15.30
6	3.417	162	14.58	5.528	136	12.24
7	4.908	148	13.32	7.462	116	10.44
8	6.673	126	11.34	10.581	53	4.77
9	9.312	88	7.92	23.778	18	1.62
10	23.778	24	2.16	136.945	6	0.54

TABLE IV: Clustering by $k-$Means with $k = 10$

According to the table IV, the result of median method is better than mean one. This table shows the θ value mainly distributed in range $[-2, 7]$. The maximum θ 136.495 represented for all-true answers (50/50 questions). Number of students who answer 49/50 questions is 18 (occupied 1.62%), or 9.8 point in raw score (could be round to 10). We have 2.16% for the highest ranking group after totaling these numbers. This rate is reasonable in many standard grading systems.

This graph shows the comparison between two clustering distribution methods:

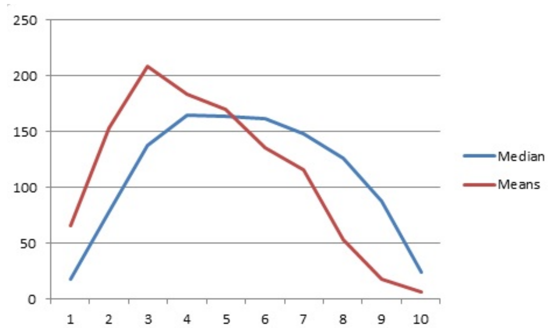

Fig. 3: Clustering distributions of mean and median

This distribution has a shape of standard distribution [10, page 24] (Gaussian distribution). According to assessment theory, this is a optimal distribution in pupil grading. Referencing to American pupil grading, the [A, B, C, D, F] scale corresponds to our [0, 10] system, we have a correlation about grading scales.

Number	**Letter**	**Ref Rate**	**Median**	**Means**
9-10	A	10	10.08	2.16
7-8	B	20	24.66	15.21
5-6	C	30	29.34	27.54
4	D	20	14.85	16.56
0-3	F	20	21.06	38.52

TABLE V: Grading systems

According to table V, the curve issued from median method looked more similar to standard distribution than mean method one. Thus, the classifying result is better.

We extracted 20 from 1111 students to sample and compare between IRT and CTT methods:

StudentID	CTT	Rank CTT	IRT	Rank IRT	Diff
11	50	1	136.9454	1	0
17	49	2	12.5534	2	0
19	49	2	9.7655	4	-2
8	48	4	12.0505	3	1
12	47	5	7.2623	6	-1
3	46	6	6.5428	9	-3
10	46	6	5.7727	10	-4
14	46	6	6.6558	8	-2
18	46	6	7.4833	5	1
6	45	10	6.9573	7	3
9	45	10	5.3995	11	-1
5	43	12	3.9266	12	0
16	42	13	3.7223	13	0
13	40	14	3.288	15	-1
15	40	14	3.0413	16	-2
20	40	14	3.6061	14	0
4	39	17	2.8044	17	0
2	33	18	1.3124	19	-1
1	31	19	0.643	20	-1
7	31	19	1.5193	18	1

TABLE VI: Comparison of CTT and IRT

As the result from table VI, the difference of two assessment systems is not too large, the fact that who answers question in right way may get better result. Though, IRT created smoother and more detailed results comparing to CTT method.

VI. Conclusion and future work

This study proposes a system of student ability assessment based on item response theory and applied $k-$Means in clustering students into pre-defined levels. Experiment results confirmed that the proposed method can produce more precise student level distribution and classify student ability more acurately. For the next step, we can apply some of machine learning methods for better result and larger input data, We also find the ways to reduce the errors of $k-$Means (outlier problem) or local minima of this algorithm [17, page 199]. Beside these problems, the selected method deeply depends on choosing initial centroids. To solve this, we can use some neural networks to deploy $k-$Means with some improvements in updating weighted matrix. We also hope to apply other comparative leaning algorithms like SOM (Self-Organising Feature Map) [13] for fastening computing speed.

Accurate estimation could help test makers to adjust testing difficult level to classify more and more exactly. Through this process, they can analyses and modify each of items to avoid standard deviation or noise. In Computerize Adaptive Test (CAT), applying item response theory helps e-learning systems suggest learning documentation for a better achievement. In addition, it could help experts to build online testing or other intelligent tutoring systems.

This result can be applied to integrate with collaborative learning system, or intelligent tutoring system or adaptive E-learning systems.

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